Postgraduate Workshop on AI Applications in Humanities

Workshop 2
Web Scraping

Jiaxuan Yu, Dian Jiang
Department of English, CUHK

The conference/seminar has been supported by the Postgraduate Students Conference/SeminarGrants of the Research Grants Council, Hong Kong.

Agenda

1. Background Information
2. Tools & Setup
3. Hands-on Activity 1: Step-by-Step Scraping Tutorial

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Part 1 Background Information

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How to deal with the data?

- Work with whatever format the website provides?
- Copy and paste information manually into a new document?

Web Scraping

So……

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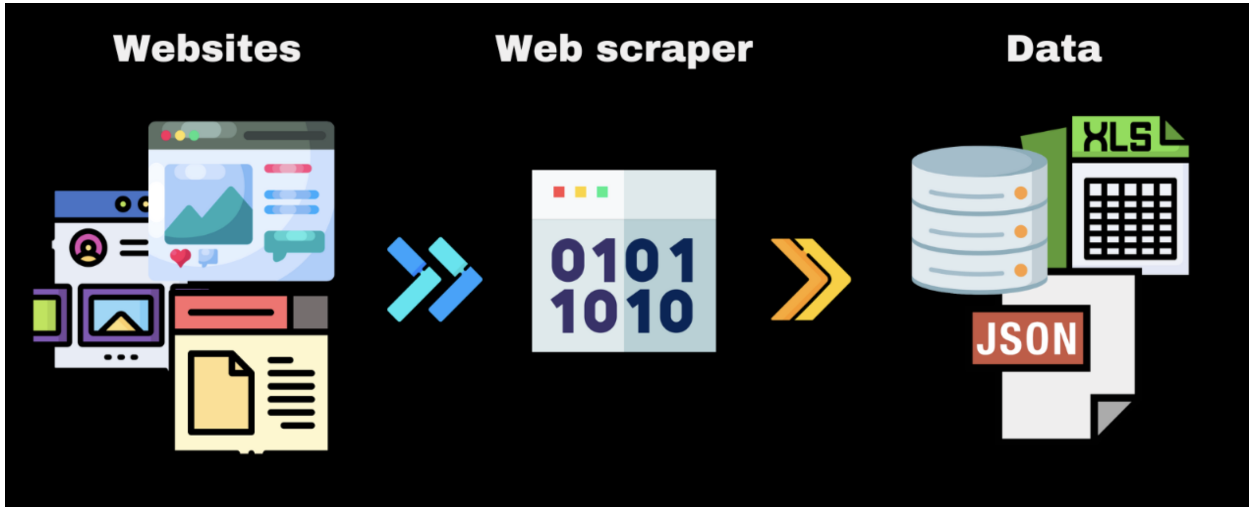
What is Web Scraping?



- Web scraping: The automated process of extracting data from a website
- Other expressions: web harvesting, data extraction, web data extraction, data scraping, data mining

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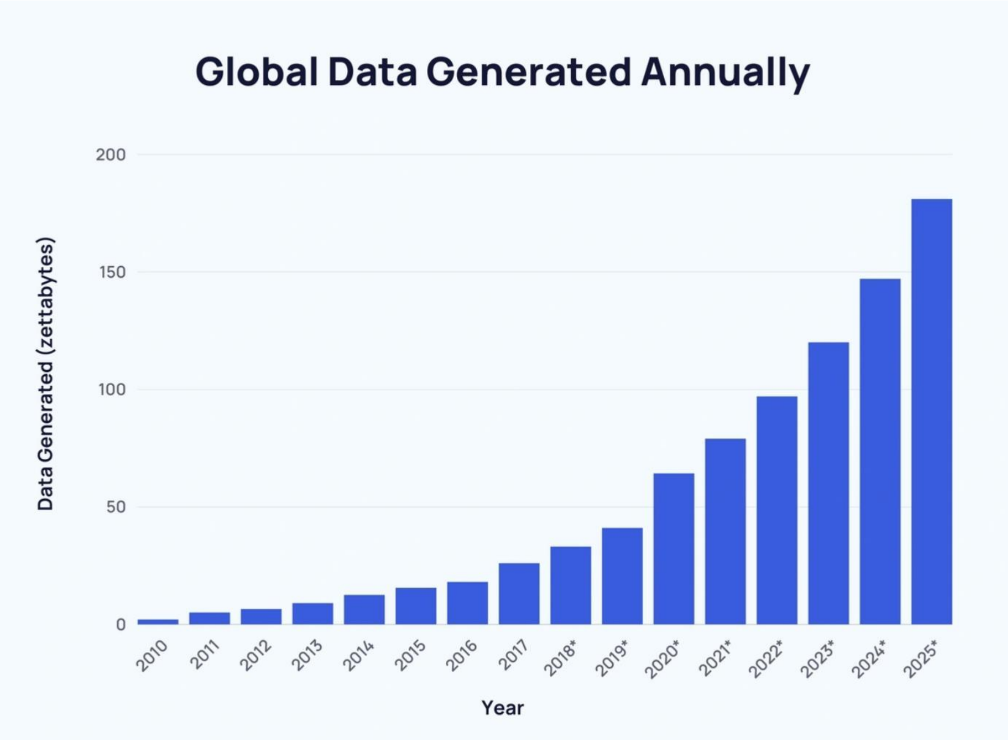
Why we use web scraping?

- The amount of data created daily in 2025: 402.74 million terabytes



It's already impossible to process even a fraction of the data on the web!

We need machines to rapidly read that data for us ☺

Duarte, F. (2025, April 24). Amount of data created daily (2025). *Exploding Topics*. https://explodingtopics.com/blog/data-generated-per-day.

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What is scraping data?

- Getting juicy unstructured data and transforming it into something more useful.

- Examples of Unstructured data

  e.g., text, images, prices, contact details, any other information publicly displayed on a web page

- Unstructured data --------------------> Structured data
  (scraping)

stored in a database or file for further analysis or use

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已由@vibemusic.置顶
@vibemusic. 3周前
Where is everyone listening from? 🌎
7 回复
7 条回复

@michaelpaz9340 3周前
Desde Lima-Perú, saludos a todos !
翻译成英语
回复

@TestTest-pf9wd 3周前
Vibe Musik🩵🩵🩵🩵🩵🩵🩵🩵🩵
翻译成英语
3 回复

@yunme730ar 3周前
NYC
2 回复

@chasleycupido 3周前
South Africa
3 回复

@NokseraDSangma-o2f 2天前
❤️
回复

@akho_crz 3周前
India- northeast❤️
回复

@rachelwilson6266 3周前
🌬️💙🔧🌻🚔
3 回复

@TakyLee28 3周前
it's 67 likes now
1 回复

@Smh.J4y 3周前（修改过）
First
3 回复

@lucsemmovies7606 3周前
Van I have a like?
4 回复

@JAYDENEMMANUELJOSEPHMoe 3周前
WHATCU TALKIN ABT MAN 🥀🥀🥀🥀🥀🥀💔💔💔💔💔💔
1 回复

@NokseraDSangma-o2f 2天前
@JAYDENEMMANUELJOSEPHMoe lmao😂
回复

隐藏回复

@AliAkbar-fv3lv 2周前
p
回复

"Reader Version"

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	cid	text	time	author	channel	votes	replies	photo	heart	reply	time_parsed	video_url
0	UgwH5PtQWLAYpeKCABB4AaABAg	Where is everyone listening from? 🌎	1 dag geleden	@vibemusic.	UChO8h2G8UjOVc081rgYU8XQ	6	6	https://yt3.ggpht.com/ytc/AIdro_korvXCmjDqO5uJ...	False	False	1.769664e+09	https://www.youtube.com/watch?v=48Jkro0IN3A
1	Ugx8tl0ri6p4KxgmVQB4AaABAg	NYC	1 dag geleden	@yunme730ar	UC_z7SKGbBmXTgXJkJjXiBtw	2		https://yt3.ggpht.com/ytc/AIdro_kEPEMtg9a1TFAX...	False	False	1.769664e+09	https://www.youtube.com/watch?v=48Jkro0IN3A
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5	UgxOv7VwmUI-bWcKNWJ4AaABAg	South Africa	1 dag geleden	@chasleycupido	UCPtZp45abHsS_VrMBwZDyww	3		https://yt3.ggpht.com/ytc/AIdro_kr4bNGofmTMPp5...	False	False	1.769664e+09	https://www.youtube.com/watch?v=48Jkro0IN3A
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"Analyst Version"

What are the benefits of web scraping?

- Fast and efficient
- Data extraction at scale
- Cost-effective and flexible
- Reliable and robust performance
- Low maintenance costs
- Delivers structured data

… …

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What is web scraping used for?

Some Real-world Use Cases

- Market research
  - Collect product names, prices, and customer reviews
- Healthcare data extraction
  - Collect data on medical research, patient reviews, etc.
- Sentiment Analysis
  - Collect opinions, reviews and comments from social media, forums, and review sites
- Academic research
  - Collect data from publicly available sources for scientific studies and analysis

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Using Web Scraping in Academic research

- Sociology: studying public opinion on social, political, or cultural topics
- Media studies: analyze audience response in the Entertainment industry
- Linguistics: examine digital communication trends in education settings (e.g., emoji use, abbreviations)
- Political Science: track public opinions on policies or political figures

… …

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Part 2 Tools & Setup

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This section introduces some common symbols and expressions you will see in Python code.

How Python speaks?

Don’ t worry about memorizing everything — this is just to help you feel less confused when we start.

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```python
## Cell 1

## Installing and loading packages

!pip install youtube-comment-downloader

import os
import json
import pandas as pd
os.chdir("/content/")
```

```python
## Cell 3

#   Initialize an empty list to store all information
all_data = []


#   Create a loop for each url
for i, url in enumerate(urls):

    # Name JSON
    json_file = f"YoutubeComments_{i}.json"

    # Use youtube-comment-downloader
    os.system(f"youtube-comment-downloader --url {url} --output {json_file}")

    # Read JSON
    with open(json_file, "r", encoding="utf-8") as f:
        json_data = []
        for line in f:
```

?

```python
#   Concatenate all DataFrames into one
combined_df = pd.concat(all_data, ignore_index=True)

#   Save the combined DataFrame to a single CSV file
combined_df.to_csv("youtube_comments.csv", index=False, encoding="utf-8-sig")

#   See how "combined_df" looks
combined_df
```

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Copy of youtube_comment_crawling.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Connect Share

[ ]

```
## Cell 1

## Installing and loading packages

!pip install youtube-comment-downloader

import os
import json
import pandas as pd
os.chdir("/content/")
```

Show hidden output

[ ]

```
## Cell 2

## Add your own Youtube link(s)
urls = [
    'https://www.youtube.com/watch?v=48Jkro0IN3A',
]

```

[ ]

```
##  Cell 3

#   Initialize an empty list to store all information
all_data = []


#   Create a loop for each url
for i, url in enumerate(urls):

    # Name JSON
    json_file = f"YoutubeComments_{i}.json"
```

IMPORTING LIBRARIES

STRINGS "" OR ''

ASSIGNMENT (=)

LISTS []

INDEXING (0,1,2...)

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IMPORTING LIBRARIES

```
Python

import pandas as pd
```

`import` = bring a tool into our workspace.

`pandas` = a data analysis library (established code package)

`as` **`pd`** = gives it a short nickname (pd)

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STRING "" OR ''

```
Python

"YoutubeComments.json"
```

Quotation marks mean the content inside is **text (a string)**.

Python treats text differently from numbers.

"5" is text
5 is a number

ps: Python doesn' t care whether you use single quotes or double quotes—
you choose the one that makes your text easier to write.

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ASSIGNMENT (=)

```python
json_file = "YoutubeComments.json"
```

= means "assign".

It stores a value on the right into a variable on the left.

Think of it as: **put this value into this box**.

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LISTS []

```
Python

urls = ["url1", "url2", "url3"]
```

Square brackets [ ] create a **list**.

A list is a collection of items.

Lists can store different types of data, such as numbers, text, or web links.

Items are separated by commas.

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INDEXING (0,1,2...)

```python
Python

data = pd.DataFrame({
    'name': ['Alice', 'Bob', 'Charlie'],
    'age': [25, 30, 35]
})
```

Python starts counting from **0**, not 1.

[0] means the first item.

[1] means the second item.

	name	age
0	Alice	25
1	Bob	30
2	Charlie	35

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Part 3 Hands-on Activity 1: Step-by-Step Scraping Tutorial

Click here to access the code

https://tinyurl.com/4zsymu7f

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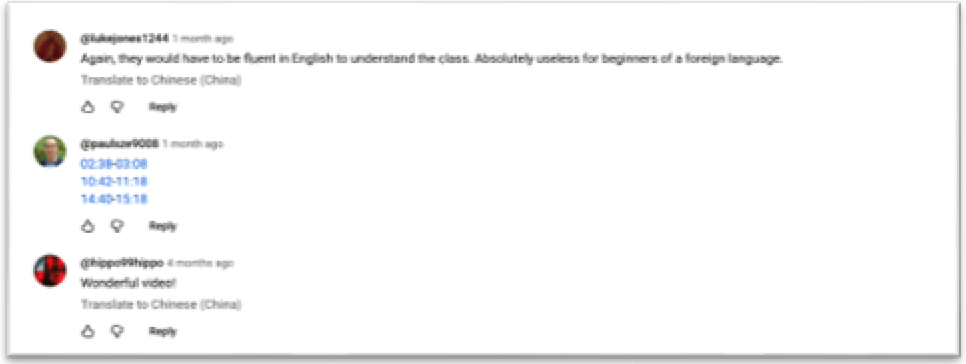
Language Teaching Methods:
Communicative Approach VS **Suggestopedia**



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Summary

Cell 1: Set up and prepare for a web scraping project

(1) Install necessary tools (youtube-comment-downloader)

(2) Import required Python library

(3) Set working directory: Ensures all file operations happen in /content/

Cell 2: Tell the program which YouTube videos you want to analyze

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Summary

Cell 3: the main web scraping cell that downloads the YouTube comments, processes them, and combines everything into a single dataset

(1) Loops through each video in your URLs list
(2) Downloads comments using youtube-comment-downloader
(3) Reads the JSON file and converts each line to a dictionary
(4) Creates a DataFrame from the comments
(5) Adds the video URL as a column to identify the source
(6) Saves the DataFrame to a list
(7) Deletes the temporary JSON file
(8) Combines all DataFrames into one master DataFrame called combined_df

Cell 4: Save the results and view the data

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